Dong 2020 Prostate — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant cells from all 4 patients (9,698; ≥100-cell floor, no sample dropped) form one continuous manifold — 0 detached islands, single connected component; every transcriptional lobe contains all 4 patients. Judged on the embedding, not the composition-capped iLISI (1.78/4, one patient ≈49% of cells).

Gate 2 — module geography → GO (non-branching)

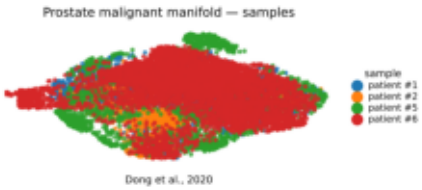
Single developmental axis: Gast & pSom co-vary (smoothed r=+0.90; joint-high enrichment 2.4 — an overlap, not competing arms) and jointly oppose neural-crest (pSom↔Crest r=−0.53). Along pseudotime Gast/pSom decline (ρ −0.58/−0.71), Crest rises to a terminal end (ρ +0.79, Crest-high decile pt 0.98). Non-branching — passes.

Gate 3 — hub cross-repression → NA

No bifurcation (fold-free single curve, 2 tips / 0 forks). The branching-only cross-repression test does not apply; correlation & modules notebooks are not produced.

STEP FIGURES

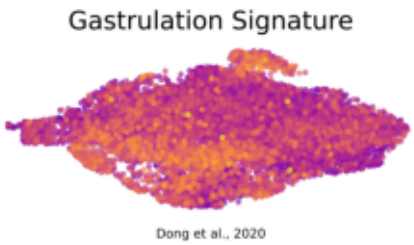
Manifold — samples



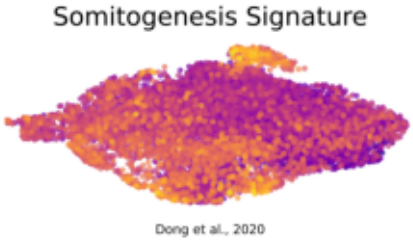
Pseudotime



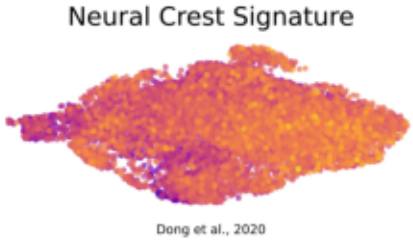
Gastrulation



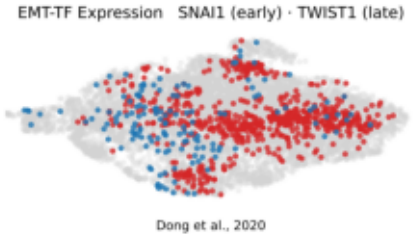
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Prostate adenocarcinoma (Dong et al. 2020), 10x — 9,698 malignant cells across 4 patients (≥100-cell floor, all samples kept; publication cell_type calls). Non-branching embryonic axis.

Verdict.
The embryonic program reproduces as a single monotone gradient — a Gast/pSom-high (overlap) origin flowing to an inverse neural-crest terminus — the collapsed-bifurcation (Linear-pSom/Crest) case, distinct from Puram's hub-and-spoke. Gate 1 GO, Gate 2 GO, Gate 3 NA.

Parameter choices

Malignant / threshold	9,698 cells, 4 patients; ≥100 floor (no cliff; all kept)
Integration	Harmony on sample, θ=10 (one patient ≈49%); spread=1.3
Compartments	Leiden res=0.15 → Body 8,696 + Cycling 1,002; whole kept
Trajectory	scFates curve N=20 λ=0.3 μ=100; 2 tips 0 forks (self-approach 2.59)
EMT-TFs	SNAI1 (root, 6.0%) → TWIST1 (terminus, 26.8%); Δpt=0.37
Overdispersion / PCA	1,670 overdispersed; 50 PC → 20 for kNN
Island removal	radius graph 6× median NN → 0 removed (one component)
Cell-cycle	cc-free (anti-corr survives 7,988 non-cycling; pt↔cyc ρ=−0.24)
Root / Orientation	tip 2 (Gast/pSom-high); rigid rotation −170.4°, corr(x_lin,pt)=+0.85